

# Deterministic Intelligence and Governance of Probabilistic Multi-Agent Systems: From Detection to Verifiable Enforcement

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*A policy-bound governance state can travel across cooperating agents without losing version awareness or audit lineage.*

Figure 1. The Verifiable AI Control Plane. A seven-stage sequence connecting a detected condition to policy selection, rule evaluation, an enforceable machine state, executed control, and preserved evidence.

The recently introduced federal proposal formally titled the “AI Kill Switch Act” raises a broader engineering question: how can AI intervention be made proportionate, enforceable, traceable, and

independently verifiable?

Representatives Ted Lieu and Nathaniel Moran announced the bipartisan proposal on July 23, 2026. Although often described in binary terms, the proposal contemplates a wider set of technical controls: stopping inference, terminating or suspending access, restricting particular capabilities, changing inference rates or compute allocation, and transitioning an affected operation to a backup system or earlier model version.<sup>1</sup>

That range matters. The engineering challenge is not simply whether a model can be stopped. It is how a control plane determines what happened, which policy applies, what may continue, what must be restricted, which component must enforce the decision, and what evidence demonstrates that enforcement occurred.

## **The Policy Question Is an Implementation Question**

The proposed framework would require covered entities to maintain core technical capabilities while directing the Secretary of Homeland Security to consider graduated deployment corrections calibrated to the severity and immediacy of a risk. The contemplated measures extend from throttling and capability restriction to suspension, shutdown, and transition to a backup system or earlier version. The text also requires consideration of whether an intervention could disrupt critical infrastructure.<sup>1</sup>

The draft's initial definitions are limited rather than universal. Subject to annual rulemaking, it uses a gross-revenue threshold of at least \$500 million for a covered entity and a development-compute-cost threshold

exceeding \$100 million for covered technology. It excludes entities that operate or make the relevant technology available solely for personal, academic, or noncommercial use.<sup>1</sup>

Following an emergency order, the framework would require preservation of model weights and telemetry, notice where practicable, confirmation that the ordered action was carried out, and governmental verification through audit, telemetry, on-site inspection, or another form of forensic review.<sup>1</sup>

Policy cannot implement itself. These requirements depend on software and infrastructure capable of translating a detected condition into an enforceable machine state and then demonstrating that the selected state was carried out.

## The Missing Control Plane

A verifiable intervention framework must answer at least five operational questions:

1. What condition occurred?
2. Which policy and policy version governed it?
3. What output, continuation, tool, workflow, or external-action state was selected?
4. Which component enforced that state?
5. What evidence demonstrates that enforcement occurred?

**Figure 1. The Verifiable AI Control Plane.** A seven-stage control sequence connecting a detected condition to policy selection, rule evaluation, an enforceable machine state, executed control, and preserved evidence.

# Where the Determinism Levels Enter

The federal proposal does not use the six-level taxonomy. The framework is applied here as an independent technical crosswalk for distinguishing different forms of repeatability, invariance, and control.

*A Determinism Level is the highest layer demonstrated under declared conditions — not a claim of universal determinism.*

- **DL1 — Code:** fixed code-level functions and evaluations under declared inputs and environment.
- **DL2 — Workflow:** repeatable routing, sequencing, and decision rules.
- **DL3 — Artifact:** fixed artifacts, versioned records, hashes, and integrity evidence.
- **DL4 — Semantic:** preservation of controlling meaning across implementation contexts.
- **DL5 — Control:** preservation of authority, constraints, tool permissions, continuation boundaries, prohibitions, and escalation rules.
- **DL6 — Release:** whether the same declared artifact and external state produce the same authorized release, restriction, rollback, or other disposition.

The control plane becomes especially significant at DL4 through DL6. At those levels, implementation may change while governing meaning, authority boundaries, and release conditions must remain intact.

The taxonomy does not establish that every artifact has demonstrated every level. It separates the level actually demonstrated from broader claims the evidence does not support.

The Grounded DI architectures discussed below address different portions of the control plane. **Entropy Governance** concerns policy-bound classification, metric selection, threshold evaluation, constraint paths, and machine-actionable dispositions. A versioned policy snapshot may identify enabled measurements, applicable thresholds, escalation mappings, permitted tool states, review requirements, and audit requirements. Binding that version to the evaluated event records which rules were in force when the decision was made.<sup>3</sup>

The **Entropy-Linked Override Chain, or ELOC**, addresses what happens after a configured deviation or threshold condition is detected. It describes a framework for activating a graduated intervention, selecting an approved constraint path, assigning a disposition, and recording the resulting control event.<sup>2</sup>

**AGDI**, as used in the referenced materials, concerns state-bound control over what an agent may do next. Its control states may govern tool invocation, workflow routing, role transitions, downstream calls, external transfers, and further execution.<sup>4</sup>

**Output-State Control** addresses a different boundary: whether a candidate artifact may become validated, final, usable, transmitted, or externally releasable. A candidate may remain in an unvalidated controlled state until its required rule, domain, source, audit, version, and integrity conditions have been satisfied.<sup>5</sup>

A governance layer complements rather than replaces **provider infrastructure**. Broader account, endpoint, networking, compute, storage, telemetry, deployment, model-weight preservation, and restoration actions still depend on the systems connected to the applicable control interface.

These functions may occur sequentially, in parallel, repeatedly, or at different execution boundaries. They are neither isolated products nor a mandatory rigid pipeline.

## **Detection Must Distinguish Signal From Activation**

The referenced detection flow begins with a monitored output or machine state. That state enters a detection layer, which produces a deviation value or condition. A comparator then evaluates the result against configured deterministic threshold conditions.

The disclosed flow contains two paths. When the threshold is not satisfied, the system continues monitoring without activating an intervention. When the threshold is satisfied, the next control stage is activated.

That negative branch is essential. A graduated control architecture should not treat every irregularity as an emergency. It must distinguish an observed condition from an actionable condition and preserve ordinary operation when the governing activation criteria have not been met.

In the disclosed architectures, entropy functions as a configured measure of detectable deviation within a controlled machine process. Depending on the implementation, the evaluated condition may concern logic-path deviation, prompt drift, state-integrity deviation, output instability, unauthorized tool behavior, external-action deviation, or multi-agent propagation.<sup>2</sup>

The measurement acquires operational significance through its policy

context: the selected metric, configured threshold, relevant domain, evaluated state, and resulting control path.

## **From Activation to Proportionate Response**

The disclosed override structure shows that activation need not produce one fixed terminal action. A threshold event may enter a multi-stage process capable of selecting among different responses according to the applicable condition, severity, domain, state, or approved path.

Possible dispositions described in the referenced materials include continued auditing, reinforcement, correction, reversion, reseeding, human review, sandboxing, restriction, suppression, denial, blocking, and audit locking. These are selectable responses rather than a universal ladder that every implementation must execute in the same order.<sup>2</sup>

This supports proportionality. A localized tool violation may justify denying a particular action without disabling unrelated functions. A compromised workflow might be redirected to a sandbox. An unstable output might be withheld or regenerated. A more serious condition might require rollback, suspension, or escalation to provider infrastructure.

The design therefore separates two decisions that are often collapsed:

- Has an actionable condition occurred?
- What response is authorized for that condition?

## **Constraint Trees Turn Policy Into Executable**

# Paths

A policy becomes operational when it can govern a machine decision.

The referenced constraint-tree architecture connects an enforcement layer to a defined constraint structure capable of identifying permitted paths, nonconforming paths, escalation points, and resulting dispositions.

In a traceable implementation, the constraint tree is versioned, its nodes and paths are identifiable, its evaluation is reproducible under the applicable configuration, and its result is available in machine-readable form.

Those properties make it possible to reconstruct which tree version was selected, which node was evaluated, which branch was taken, which condition caused that branch, which disposition followed, and whether the same path existed under a prior policy version.

Without that structure, a system may impose an intervention without being able to reconstruct why it selected that response.

## State-Bound Agent Continuation

The referenced state-bound escalation architecture begins with an agent-operation request and classifies the requested operation across several conditions. The illustrated categories include the agent, task, role, domain, risk, tool use, output class, and workflow context.

The classification output can select a versioned constraint structure and identify the relevant node path. A detected violation may then be mapped through a sequence such as:

## **Violation State → Escalation Mapping → Escalation State → Control-Action Mapping → Machine-Readable Action**

The illustrated actions include fallback, correction, suppression, sandboxing, restriction or rerouting, and halt or allowance. Other disclosed configurations may use delay, rollback, review, or escalation.

The important feature is that the result is not merely an explanation addressed to a user. It is a state that can be consumed by an execution controller.

The continuation architecture identifies several pre-continuation boundaries:

- tool invocation;
- workflow routing;
- downstream computer-system calls;
- role-transition operations; and
- further execution.

A system may therefore permit an agent to prepare an internal plan while preventing it from invoking a live tool, modifying an external record, transferring data, changing roles, or initiating a public workflow. Control attaches to the operation being attempted rather than to the agent as an undifferentiated whole.

This is where DL5 becomes distinct from lower levels. The relevant question is no longer only whether the same code or workflow runs. It is whether authority boundaries, prohibitions, continuation gates, and escalation rules survive autonomous execution intact.

## **Output-State Control**

Agent continuation and output validity are related but distinct questions.

An agent may follow an authorized workflow and still produce an artifact that lacks required sources, contains an unresolved procedural defect, uses the wrong policy version, or has an incomplete audit record.

The disclosed Output-State Control architecture can retain that artifact in an unvalidated controlled state while evaluating rule constraints, domain restrictions, source status, logic-chain alignment, certainty indicators, version consistency, audit completeness, and integrity conditions. The candidate transitions to a validated state only after the applicable requirements are satisfied. Otherwise, it may be withheld, rejected, quarantined, suppressed, rerouted, revised, regenerated, or reprocessed.<sup>5</sup>

This is more than a warning that an output may be unreliable. It governs whether the artifact can become final, usable, stored, transmitted, or released.

That distinction connects DL3 to DL6. DL3 may establish the identity or integrity of an artifact. DL6 concerns the separate question of whether the declared artifact and applicable state produce the authorized release or non-release disposition.

## **Governance Becomes Real at the Execution Boundary**

The referenced tool-use control flow illustrates the point at which policy becomes machine control.

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A tool-use or external-action request is evaluated against an approved deterministic logic path. A conforming request reaches an allowed execution path. A nonconforming request may receive a blocked, denied, sandboxed, restricted, or audit-locked disposition.

That distinction is material. A prose warning can be ignored by a downstream component. A machine-readable disposition can be consumed by a tool-access controller, workflow gateway, API gateway, middleware layer, orchestration service, execution-state controller, or native agent runtime.

The consuming component may then disable the requested tool, prevent workflow routing, stop a downstream call, restrict a role transition, redirect the operation, or halt continuation.<sup>4</sup>

A control system is therefore incomplete if it merely detects a problematic condition. It must connect the decision to the boundary at which the prohibited operation would otherwise occur.

## Control That Propagates Across Agents

Single-agent enforcement is insufficient when an operation is distributed across multiple agents, tools, or orchestration layers.

The referenced disclosure describes a multi-agent architecture in which a control state may propagate across cooperating agents while

preserving separate constraint evaluation, version awareness, and linked audit lineage.

In that architecture, multiple generative agents operate through respective mirrored constraint trees. A shared state broker or inter-agent interface exchanges policy-controlled state information among them. That information may include rule and version identifiers, node-path identifiers, state hashes, lineage pointers, audit identifiers, threshold-condition identifiers, override-tier identifiers, and disposition identifiers.

A deviation detected in one agent produces a propagated override state. Connected agents can receive or evaluate that state through their own mirrored constraint structures. Downstream activity may then be sandboxed, blocked, restricted, or audit-locked according to the applicable policy and local execution boundary.

The mirrored trees matter because propagation need not erase agent-specific evaluation. A shared intervention state can travel across the environment while each receiving agent remains subject to its own defined constraint structure.

Linked metadata connects the originating condition to the propagated response. That permits later review of where the condition arose, which state was transmitted, which agents received it, which rule versions they applied, and what downstream dispositions followed.

Control need not be isolated to a single model instance. A policy-bound governance state can travel across cooperating agents without losing version awareness or audit lineage.

The significance is not that every agent must be disabled together. It is

that a control decision can become portable, state-bound, version-aware, and traceable across a multi-agent environment.

That is also where the distinction between DL4 and DL5 becomes useful. The controlling meaning of the intervention must remain stable as it moves between agents, while each receiving system must preserve its own authority boundaries and constraint structure.

## **When Technical Terms Become Testable Architecture**

Patent figures and written disclosures do not, by themselves, establish empirical performance, independent safety certification, patent validity, claim coverage, statutory compliance, or production readiness.

They do support a narrower but meaningful form of architectural validation: central terms can be mapped to identifiable machine functions, inputs, evaluated conditions, state transitions, enforcement points, interfaces, dispositions, and audit records.

A technically meaningful term should permit concrete questions:

- What enters the component?
- What condition does it evaluate?
- Which policy or version governs the evaluation?
- What machine state does it produce?
- Which execution boundary consumes that state?
- What occurs when the condition is not satisfied?
- What occurs when it is satisfied?
- What evidence records the result?

The referenced disclosures permit those questions to be asked across detection, threshold comparison, override activation, constraint-tree evaluation, continuation gating, tool-use control, multi-agent propagation, and audit recording.

The value of those disclosures is not that they prove an implementation succeeds. Their value is that they define concrete machine functions that can be independently reviewed, questioned, tested, measured, reproduced, or challenged.

That is architectural intelligibility, not empirical or legal certification.

## **Audit Lineage: Evidence, Not Merely Explanation**

An enforcement system should preserve more than a narrative log stating that an action occurred.

The referenced audit-record structure contains identifiable fields. In a broader implementation, an integrity-protected record may link policy identifiers and versions, threshold conditions, evaluated node paths, state identifiers, timestamps, hashes, prior-state references, control actions, dispositions, enforcement status, lineage references, and related machine-readable metadata.

Those records can support reconstruction of what condition was evaluated, which policy governed it, which path was taken, what response was selected, whether the corresponding enforcement action occurred, and whether the preserved record indicates later alteration.

The applications also describe records linking agent and task conditions, violation and escalation states, continuation states, output

states, hashes, timestamps, prior-state references, and dispositions.<sup>2345</sup>

An internal record does not automatically satisfy every legal requirement for model-weight preservation, infrastructure telemetry, evidence retention, notice, regulatory reporting, or forensic inspection. It can, however, provide a structured source of evidence for those broader processes.

This is the role of DL3 within the control plane: establishing fixed, inspectable evidence of the artifact, policy version, decision path, or enforcement event. It does not, by itself, establish that the underlying decision was correct.

## **Provider Infrastructure and Operational Continuity**

A governance layer complements rather than replaces provider infrastructure.

System-wide actions — including termination of inference, production-endpoint shutdown, compute reallocation, account-wide suspension, network isolation, storage preservation, model-weight retention, and restoration of a prior deployed version — depend on provider identity, orchestration, networking, compute, storage, telemetry, and deployment systems.

The governance architecture can identify the applicable condition, select a disposition, produce an enforceable state, and issue or enforce a control instruction through the available interface. Connected infrastructure must perform the corresponding platform action.<sup>2345</sup>

This division also matters for critical-infrastructure continuity. A platform-wide intervention could affect healthcare, communications, water systems, transportation, energy, emergency response, financial services, or other dependent systems. The federal proposal expressly directs attention to that risk and includes transition to a backup system or earlier version among the graduated measures to be considered.<sup>1</sup>

Fallback routing, capability-specific restriction, sandboxing, account-level suspension, and rollback may permit narrower containment where the circumstances support it. Full system shutdown remains a distinct response for conditions that justify it.

## **Controlled Testing Before Emergency Use**

An emergency control capability should be testable before an emergency occurs.

The disclosed architectures include mechanisms that could support simulated triggers, constrained tool use, sandboxing, rollback, review routing, lineage recording, and audit verification. The federal proposal separately defines red-teaming as structured adversarial testing conducted in a controlled environment that simulates real-world conditions.<sup>1</sup>

A controlled test could examine whether:

- the expected policy version loads;
- the correct threshold condition activates;
- the intended constraint path is selected;
- the continuation gate blocks the targeted operation;
- unaffected operations remain available;

- a propagated state reaches the intended agents;
- fallback routing functions;
- the enforcement record is generated; and
- the system can return to an approved configuration.

Testing those relationships is more informative than checking only whether a terminal command exists.

The result of such testing should be expressed at the highest level actually demonstrated under the declared conditions. A successful artifact replay may support DL3 evidence. Preservation of controlling meaning may support DL4. Preserved authority and escalation boundaries may support DL5. Reproducible authorization or non-authorization under a declared artifact and external state may support DL6.

None of those findings, standing alone, establishes universal determinism.

## Conclusion

Whether a particular implementation ultimately succeeds will depend on engineering, testing, operational experience, and deployment. The broader architectural question is increasingly clear: meaningful AI intervention requires a verifiable control plane connecting observed conditions to policy-controlled machine states, enforceable execution boundaries, and audit-ready evidence.

The engineering challenge is to make each transition explicit:

**Detected Condition → Applicable Policy → Evaluated Rule →  
Selected Disposition → Enforceable State → Executed Control →**

## **Preserved Evidence**

ELOC provides a disclosed structure for converting detected deviation into graduated machine control. Entropy Governance supplies policy-bound measurement and disposition logic. AGDI supplies state-bound control over an agent's next operation. Output-State Control governs whether an artifact may become validated or releasable. Mirrored constraint structures and shared state interfaces provide a representative architecture for extending controlled responses across a multi-agent environment.

The six-level framework supplies a compact vocabulary for distinguishing what has actually been demonstrated: fixed code, repeatable workflow, fixed artifact, preserved meaning, preserved control, or reproducible release authorization.

Together with connected provider infrastructure, these systems illustrate how emergency AI intervention could operate as a defined, proportionate, documented, and reviewable process rather than an unexplained binary command.

## **Independence and Scope Notice**

The federal proposal does not reference the six-level Determinism Level framework, Protocol A, ELOC, AGDI, Entropy Governance, Output-State Control, Grounded DI LLC, or the inventor. The taxonomy is applied here as an independent technical crosswalk, and this article is independent technical analysis.

The functional comparisons presented here do not establish patent claim coverage, statutory compliance, governmental endorsement,

infringement, priority entitlement, patent validity, production effectiveness, or a requirement that any identified architecture be used to implement the legislation. Broader platform actions would still require integration with the applicable provider's account, endpoint, network, compute, storage, telemetry, deployment, and model-control infrastructure.



## About the Author

**Mark S. Weinstein** is a litigation attorney and the creator of **Protocol A**, the deterministic reasoning framework that became the foundation of **Grounded DI LLC**.

## Source Notes

1. Official public announcement dated July 23, 2026, and the circulating legislative draft. The announcement identifies Representatives Ted Lieu and Nathaniel Moran with the bipartisan proposal. The draft identifies Mr. Lieu as the introducer, leaves the H.R. number and committee referral blank, and contains the shutdown-capability, graduated-response, critical-infrastructure, preservation, verification, threshold, exemption, and red-teaming provisions discussed above. [Official announcement](#); [legislative draft](#).
2. U.S. Patent Application N°19/726,030, *Deterministic Intelligence Systems and Methods for Entropy-Linked Override Chain Enforcement in Generative Artificial Intelligence Systems*, ¶¶ 33–90.
3. U.S. Patent Application N°19/748,124, *Systems and Methods for*

*Deterministic Entropy Governance and Entropy-Linked Override Enforcement in Generative Artificial Intelligence Systems,* including ¶¶ 20–37, 52–123, and 332–353.

4. U.S. Patent Application N°19/722,955, *Deterministic Intelligence Systems and Methods for State-Bound Continuation Control of Generative Agent Operations*, including ¶¶ 28–51 and 60–129.
5. U.S. Patent Application N°19/715,156, *Deterministic Intelligence Systems and Methods for Rule-Governed, Domain-Scoped, Audit-Traceable Control of Generative Output States*, ¶¶ 33–76.